

RAILWAY INTRUDING PEDESTRIAN IMAGE SEQUENCE GENERATION METHOD WITH POSE TRANSFER TECHNIQUES

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Abstract

Pedestrians intruding detection on deep learning method is vital for the safety of high-speed railway operations. The accuracy of deep learning object recognition is significantly depended on the volume and diversity of training samples. Addressing the issue of insufficient and hard-to-obtain intruding images in high-speed railway scenes, this paper introduces a novel method for generating railway intruding pedestrian image sequences with pose transfer. This method can generate necessary intruding samples and produce the annotated files automatically for deep learning training. To tackle the challenges of pose feature extraction, a pose transfer module based on generative adversarial network is proposed; furthermore, a feature fusion module based on spatial attention mechanisms has been proposed to preserve essential pose features. Experimental results on the DeepFashion dataset demonstrate that the proposed modules significantly enhance the quality of generated pedestrian images and the accuracy of pose transfer. In the same data set, compared with the classical pose transfer algorithm, the algorithm in this paper improves the SSIM index by 5.6% and the pckh index by 2.9%. Finally, the effectiveness of the intruding image generation is also verified by the improvement of detection algorithm accuracy.

1 Introduction

Pedestrian intruding is a major hazard to the safety of railway operations. In terms of these literature^[1-3], traditional railway perimeter intrusion detection methods are categorized into contact and non-contact types. The former facing difficulties such as limited scope and high maintenance complexity; the non-contact methods are mainstream. The rise of railway foreign object detection technology with video analysis makes the alarm information visible and easy to maintain. Although video analysis-based technologies have these advantages, they still limited by speed and accuracy, especially under low light conditions^[4]. Recent advancements with deep learning detection algorithms offer great improvements of accuracy, but their efficacy is often constrained by the high dependency on substantial data, especially by the huge manually annotated work. Due to the lack of a large number of pedestrians intruding samples in railway scenes, it significantly hampers the scalability and adaptability of deep learning methods. Moreover, despite advances such as the use of deconvolution and model compression techniques which enhance detection of small objects and speed^[5], integrated into practical applications remains a challenge. Inspired by these restrictions, our research turns to Generative Adversarial Networks^[6] (GAN) to generate detailed and diverse images. Although GAN series networks have achieved stabilized and improved outputs, practical application integration is still problematic, such as the lack of realism in data generation for railway

scenarios. Our novel approach leverages pose transfer to generate image sequences that not only improve the realism but also address the critical need for adequate training datasets in high-speed railway perimeter intrusion detection. By employing advanced GANs and pose transfer techniques, this study aims to provide an efficient, scalable solution that enhances the operational safety of railway systems globally, overcoming traditional constraints of data volume and processing demands.

2. Methodology

2.1 GSGAN: A Generative Pose Transfer Network for Railway Pedestrian Image sequence

To generate pedestrian images with target poses, this paper introduces GSGAN, which creates images of pedestrians in desired poses by transferring poses from non-railway to railway scenes. Fig.1 illustrates the algorithmic framework of GSGAN, which includes both training and application phases. During the training phase, the generator G and discriminators (appearance) D_1 and (pose) D_2 are trained with DeepFashion dataset^[7]. The generator will learn to transfer pedestrian poses while ensuring that the appearance of the pedestrians remains consistent with the original images. The discriminators that evaluated the authenticity of the samples. In the application phase, the optimised generator model is used to transfer poses from the original images, generating a sequence of pedestrian

images that retain the original appearance and achieve the target poses. In response to the challenge of extracting critical features from posture images that contain only key points and key edges, without additional pixel information, we propose a pose transfer module for the generator network based on Graph Convolutional Networks^[8] (GCN). Additionally, to preserve essential pose features, a feature fusion module based on Spatial Attention (SA) has been introduced for the generator network. The GCN pose transfer module extracts features of the poses and models the relationships between the key points of the two poses. The pose features outputted are then fed into the feature fusion module, which filters the features, retaining important ones and discarding irrelevant ones. Through an upsampling layer, the features are restored to their original size, and the output is a pedestrian image with the target pose and original appearance. The network framework diagram of generator is shown in Fig.2.

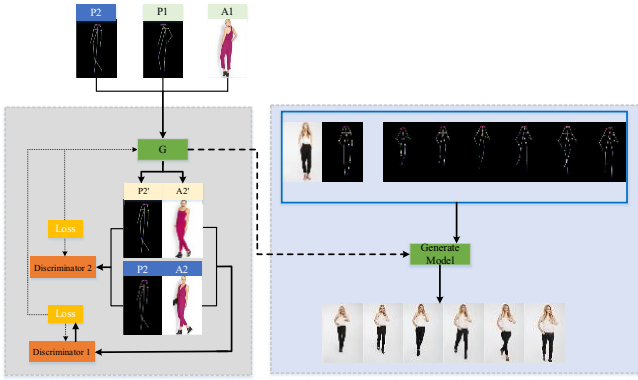


Fig. 1 Algorithm GSGAN

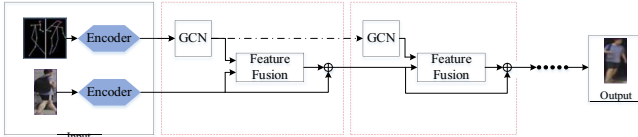


Fig. 2 Network framework diagram of generator

2.2 Graph Convolutional Network-Based Convolutional Network Module

Given that skeleton images represent an irregular graph structure, the graph consists of N nodes, each with its own features, traditional CNN cannot be applied directly to graphs. Graph Convolutional Network (GCN) can aggregate features from neighbouring nodes around a specific node, enabling feature learning and prediction through weighted aggregation. When extracting features from skeleton images. These nodes' features form an $N \times D$ dimensional feature matrix X , and the relationships between the nodes are represented in an $N \times N$ dimensional matrix A , also known as the adjacency matrix. The GCN can be operated as a neural network layer, with inter-layer broadcast as shown in Formula (1).

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} \right) \quad (1)$$

In the formula, $\tilde{A} = A + I$, I represents the identity matrix; $\tilde{D}$ is the degree matrix of $\tilde{A}$, where $\tilde{D} = \sum \tilde{A}_{ij}$; H represents the

feature matrix of all nodes at each layer, with $H^{(0)} = X$ specifically for the input layer; $W^{(l)}$ is the trainable parameter matrix for the convolutional transformation at each layer; and σ is the nonlinear activation function. As illustrated in Fig.3, the network model diagram of the pose transfer module is shown, with inputs consisting of two skeletons, P1 and P2. Initially, P2 is used to transform P1 from Euclidean space into graph space, as shown by Equation (2), producing new features. Here, an 1×1 convolution layer is employed to obtain new features in graph space, which represent the relationships between the nodes of the two skeletons. In Equation (3), another 1×1 convolution layer is used, and the resulting features are weighted at the channel level through a one-dimensional convolution. These transformed features are then added to P1 to update the target features, as indicated by Equation (4).

$$H_1 = \theta(F_0^{P2}) \quad (2)$$

$$V_1 = \varphi(F_0^{P1}) H_1 \quad (3)$$

$$\tilde{F}_0^{P1} = \phi(M_1 H_1) + F_0^{P1} \quad (4)$$

Using the GCN, the updated features of P1 and P2 are obtained and then concatenated into a feature matrix of dimensions [256, 64, 44]. This updated skeleton feature matrix not only undergoes further updates but also serves as an attention layer to enhance the appearance features.

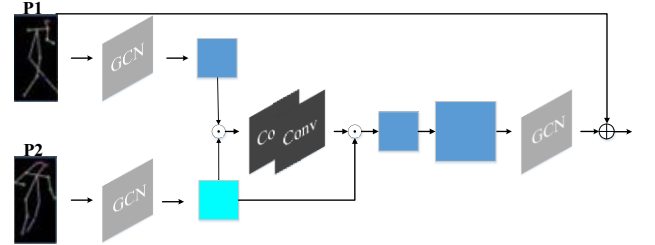


Fig. 3 Pose transfer module based on GCN

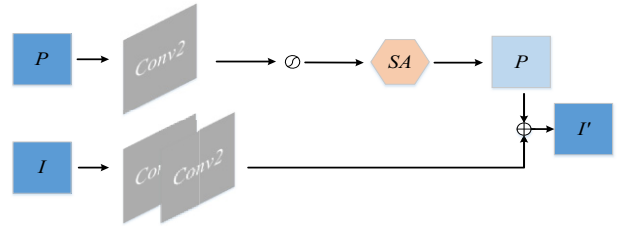


Fig. 4: Feature fusion module based on SA

2.3 Feature Fusion Module Based on Spatial Attention

To ensure the retention of crucial pose features, this paper designs a feature fusion module based on Spatial Attention (SA), as shown in Fig.4. The input to the module consists of appearance features I and pose features P . The output is the updated appearance features, enhanced by the pose features. At first, the two input features are concatenated and fed into two convolutional layers to produce an attention map A_p , as

shown in the Equation (5).

$$P' = \sigma(\text{Conv}(P'_1 + P'_2)) \quad (5)$$

where $\sigma(\bullet)$ represents the sigmoid function. To prevent overfitting due to an excessive number of parameters, this paper applies global average pooling to the feature maps of the pose, as shown in Equation (6).

$$S_c = F_{gp}(P') = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W P'(i, j) \quad (6)$$

The attention layer encodes the pose features and multiplies them element-wise with the input pose features to resume the size of its features, as shown in Equation (7).

$$M = (S_c W + b) \otimes P_c \quad (7)$$

Then the pose feature M will be dot with the appearance feature to achieve feature fusion, as shown in Equation (8).

$$I' = M \otimes I \quad (8)$$

The updated appearance and pose features are then concatenated and input into the next GCN and image fusion module. Ultimately, the generator network only needs to upsample the final appearance features back to the original image dimensions to generate pedestrian images.

3 Experiment Evaluation

3.1 Experiment Setting

This paper conducts training of GSGAN based on the DeepFashion dataset, which includes 48,674 images across three categories: upper body, lower body, and full body. The pedestrian skeleton dataset is generated via a human joint detection network, where each skeleton uniquely corresponds to an image in the dataset. For pose coordinate data, each pedestrian image name is associated with 18 X-coordinates and 18 Y-coordinates. The coordinate ranges are 0-175 for X and 0-255 for Y. As shown in Fig.5, each pedestrian image is paired with its corresponding skeletal pose. This pose coordinate data, combined with the public dataset images, forms a database for pose transfer training. The paired pedestrian poses and model images are input into GSGAN for training. For the Generative Adversarial Network, both the discriminator and generator undergo end-to-end training. The batch size is set to 16. Additionally, the Adam (Adaptive Moment Estimation) optimization algorithm is used for model parameter updates. The initial learning rate is set at 0.0002, and starting from the 499th epoch, each epoch is reduced by 1×10^{-6} based on the previous epoch, all experiments is trained through the PyTorch deep learning platform in Ubuntu 7.5.0-3ubuntu1 operating system and the computing GPU is NVIDIA Tesla V100S.



Fig. 5: Pedestrian image and her pose

3.2 Ablation Experiment

To validate the effectiveness of the proposed GCN and SA modules, this study evaluates the pedestrian images generated by the GSGAN network using three different metrics:

Inception Score^[9] (IS): This metric is used to assess the clarity of the generated images. A higher IS indicates greater image clarity.

Structural Similarity^[10] (SSIM): SSIM evaluates the effectiveness of the generated images by comparing their similarity to the target images, measuring aspects of image brightness, structure, and contrast.

Percentage of Correct Key points^[11] (Pckh): The Pckh index assesses the consistency between the generated pedestrian poses and the target poses, serving as a measure of the effectiveness of pose transfer.

The generated pedestrian image sequences were evaluated based on IS, SSIM, and Pckh metrics, with the results presented in Table 1. The original GAN network was used as a baseline for comparative analysis.

Table 1 Test results of Ablation Experiment

Method	Pckh0.5	SSIM	IS
GAN	0.9651	0.5155	1.24
GAN+GCN	0.9702	0.5137	1.24
GAN+GCN+SA	0.9938	0.5694	1.38

It was observed that the Pckh0.5 index improved from 0.9651 to 0.9702 by incorporating the GCN module, indicating that the GCN module enhanced the connection between the original and target poses, thereby improving the pose transfer effect. Furthermore, with the addition of the SA module, the SSIM and IS indices used to assess image quality increased by 0.0557 and 0.14, respectively, and the Pckh0.5 also increased by 0.0236, which can demonstrate the significant role of the SA module in enhancing both the pose transfer effect and the quality of the generated images.

3.3 Generation of Railway Pedestrian Intrusion Image Sequences

This paper collected images of empty railway scenes from various locations in the Circular Railway Test Track at the China Academy of Railway Sciences, and segments of the high-speed railways from Guangzhou to Shenzhen and from Baoji to Lanzhou. These images include different locations such as tunnel entrances, turnouts, and cuttings, as well as different weather conditions like sunny, cloudy, and rainy days.

To determine the scale of pedestrians within these railway background images, this study employed a pedestrian scale estimation algorithm based on the standard track gauge to assess the size of pedestrians at different locations within the railway scenes. Ultimately, sequences of images depicting pedestrian movement were generated across a variety of railway settings. The generated railway pedestrian intruding image sequences with different clothes in different sections such as cuttings, tunnel entrances, and turnouts are displayed as shown in Fig. 6.

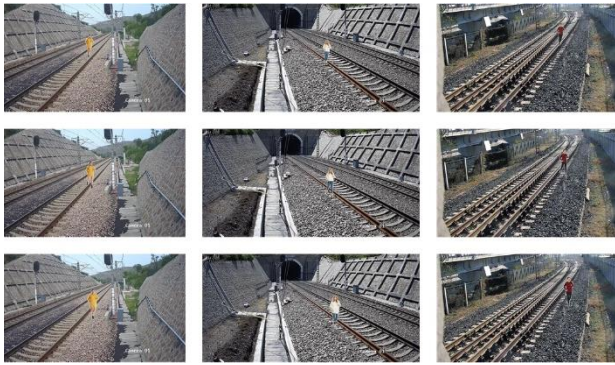


Fig. 6: Generated intruding pedestrian sequences in different railway scenes

This method can also be used to generate annotated dataset in producing intrusion images automatically, which saves a lot of manpower on annotation work. To verify whether the generated images meet the training samples requirements, this study conducted repeated tests on the image sequences using four widely used pretrained network models: Yolov5^[12], Faster R-CNN^[13], SSD^[14], and DSSD^[15]. The results were evaluated using the Average Precision (AP) metric, as shown in Table 2. The testing score for Yolov5 on the generated samples was 85.2%, differing by only 2.2% from the real sample database.

Table 2 Test results of generated image sequence

Method	backbone	Data source	Dataset size	AP(%)
Yolov5	Darknet-53	generated	4100	85.2
		real	672	87.4
SSD	VGG16	generated	4100	83.2
		real	672	84.0
Faster R-CNN	VGG16	generated	4100	81.9
		real	672	82.9
DSSD-321	ResNet-101	generated	4100	84.8
		real	672	86.3

The testing scores for the generated dataset were high across all detection networks, with discrepancies of 0.8% for SSD, 1.0% for Faster R-CNN, and 1.5% for DSSD when compared to the real dataset. These results indicate that the railway pedestrian intrusion images sequences generated in this study closely resemble real pedestrian intrusion images, thereby affirming the authenticity of the generated image sequences and fulfilling the requirements for training samples. For a more tangible demonstration of the detection effectiveness, we utilize Yolov5 to conduct image detection on the generated

railway pedestrian intrusion image sequences, with the results depicted in Fig. 7. Every pedestrian in each frame was detected correctly with high confidence scores around 0.9, which was almost higher than the confidence score of 0.85 in real intruding sequence. The result also indicates that the generated image sequences possess a high degree of realism.

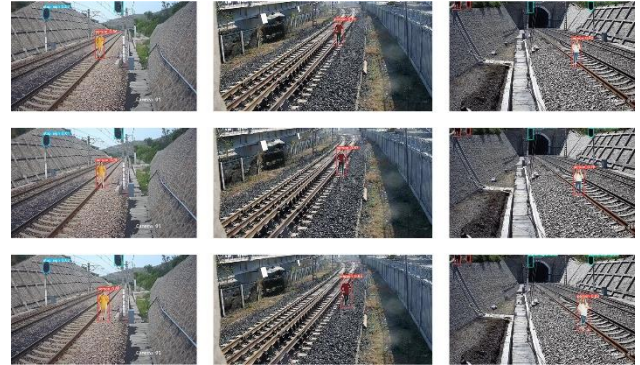


Fig. 7 Detection results of generated image sequence

3.4 Evaluating the Efficacy of Generated Images as Training Data

In order to verify the enhancement effect of the generated data on the intrusion detection network, this study created datasets including sequences with real and different number of generated samples, organized into five groups as shown in Table 3.

Table 3 Data set setting of target detection network

Model Index	Real sample number	Generated sample number	Pedestrian Number
1	672	0	966
2	672	1000	1966
3	672	2000	2966
4	672	3000	3966
5	672	4000	4966

The first group contained only 672 real railway pedestrian intrusion images, including a total of 966 intruding pedestrians; the second group added 1,000 generated images to the first, increasing the number of intruding pedestrians by 1,000; similarly, each subsequent group added another 1,000 generated images, culminating in the fifth group with a total of 4,966 intruding pedestrians. These five groups were used to train the target detection network. Consequently, five models were generated, each tested to verify the effectiveness of the generated samples as training data. This paper utilized the Yolov5 network model and assessed it using Average Precision (AP), with results detailed in Table 4.

Table 4 Test results of different training set models

Model Index	Sample Size	AP(%)
1	672	71.6
2	672+1000	72
3	672+2000	79
4	672+3000	81
5	672+4000	84.3

Using only 672 real images, Yolov5 was trained to a satisfactory state, achieving a detection accuracy of 71.6%. However, the addition of 1,000 generated images increased the model's accuracy by 0.4%, with further improvements of 7.4%, 9.4%, and 12.7% in Models 3, 4, and 5, respectively. These results indicate that the intrusion data generated in this study significantly enhances the detection accuracy of the object detection network models. For a more intuitive understanding of how the generated data improved the models, Fig.8 displays the detection performance of Models 1 and 5 on actual intruding pedestrians. Fig.8 (a) shows the performance of Model 1, which detected only a part of the intruding pedestrians. Fig.8 (b) illustrates that the detection capability in real railway intruding image sequences of Model 5 is improved after adding 4,000 generated samples as training data. Compared with Model 1, there was an increase in the number of detected targets and confidence scores of Model 5, which demonstrates that the generated intruding samples with pose transfer have significant implications in intruding detection network training.

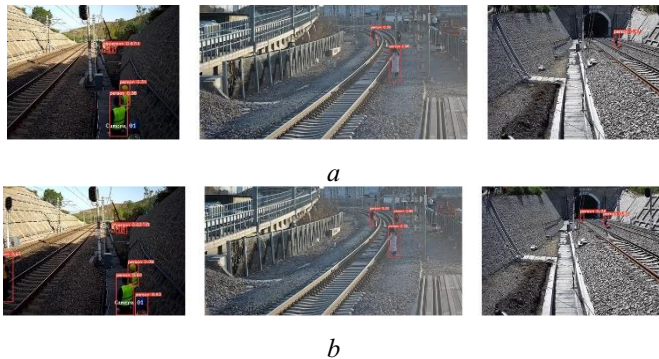


Fig. 8 Result comparison of different detection model
(a) Detection Result of Model 1
(b) Detection Result of Model 5

4 Conclusion

In response to the problem that the railway pedestrian intrusion detection network lacks pedestrian intrusion samples in railway scenes as training sets, this paper proposes a railway pedestrian intrusion image sequence generation method based on pose transfer, which transfers the posture of pedestrian from other non-railway scenes to railway scenes to generate railway pedestrian intrusion samples. Experiments show that the generated sample can be used to improve the intruding detection network performance as training data.

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6 References

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